

A Log-Linear Regression Analysis of Allometric Scaling and Species Effects on Penguin Body Mass

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Abstract

This project models radiologists' malignancy ratings of lung nodules in the LIDC-IDRI database, using geometric features computed from their outlines. We compare a proportional odds model of the median rating with a latent-grade model fitted by the EM algorithm. Size is the dominant predictor, but the proportional odds assumption is rejected. On held-out patients, the models predict exact grades poorly and the EM model does not improve on always predicting grade 3. However, a triage rule flags 91.3% of high-grade nodules while flagging 47.0% of all nodules.

All models are fitted to the trimmed file `nodules_train_trimmed.csv` rather than the entire training data: we removed individual ratings at least two grades from their nodule's median (212 of the 4647 training ratings, 4.6%, across 204 nodules) and recomputed each median. One nodule lost all of its ratings and is excluded from the naive models. The test set uses all ratings. This rule was chosen after an initial fit gave implausible results. It improves the interpretability of the latent classes, but it reduces measured disagreement, so the estimated confusion matrix overstates reader reliability, and predictive fit on the untrimmed ratings is worse.

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1 Introduction

Lung nodules are often found on chest CT scans, and most are benign. Radiologists must decide which ones need attention, and their time is limited, particularly in public health systems. A statistical model of how nodule characteristics relate to suspected malignancy could help to prioritise this review.

Modelling this is not straightforward because radiologists often disagree. In the LIDC-IDRI database, up to four radiologists rate each nodule's malignancy on an ordered scale from 1 to 5, and on about 78% of the multi-reader nodules in our training set they do not all give the same rating. The usual approach of taking one summary rating per nodule ignores this disagreement.

This project therefore compares two approaches: a proportional odds model of the median rating, and a latent-grade model, fitted by the EM algorithm, that treats a nodule's true grade as unobserved and each rating as a noisy measurement of it. We address the following research questions.

1. Which size and shape features are associated with radiologists' malignancy ratings?
2. Does the proportional odds model adequately describe that relationship?
3. How much do radiologists' semantic ratings add beyond geometric features?
4. Does modelling a latent true grade change the estimated effects or the predictions?
5. Could such a model support the triage of nodules for review?

The rest of the report is organised as follows. Section 2 describes the data, and Section 3 specifies and fits the models. Section 4 presents the hypothesis tests, including the test of the proportional odds assumption. Section 5 evaluates the models on held-out patients, and Section 6 interprets the results as a triage rule. Section 7 concludes.

2 Data Description

The dataset used in this study is the LIDC-IDRI (Lung Image Database Consortium and Image Database Resource Initiative) database[1]. It contains thoracic CT scans in which up to four radiologists outlined each lung nodule of at least 3 mm and rated its appearance and its likely malignancy.

The unit of analysis is the *nodule*. Geometric features are averaged across a nodule's readers and semantic ratings are summarised by their median. We use the median because the ratings are ordinal, so the gaps between grades have no fixed numerical meaning, and

because it is less sensitive to a single outlying reader than the mean.

The variables used in this study are as follows.

Response

- **Malignancy (`mal_median`)**: The median of the readers' ratings, on an ordered scale from 1 (highly unlikely to be malignant) to 5 (highly suspicious). A median falling between two grades is rounded up, which affects 548 nodules (20.8%). Each reader's rating is also retained for the latent-grade model. Discrete (ordinal), with support $\{1, 2, 3, 4, 5\}$.

Primary model: geometric features These are computed from the outlines drawn by the radiologists and are treated as continuous random variables.

- **Log Volume (`log_volume`)**: The natural logarithm of the nodule's volume in mm^3. The logarithm is used because volume is strongly right-skewed. Continuous, with support $\mathbb{R}$.
- **Elongation (`elongation`)**: A value near 1 means that the nodule is not elongated. Continuous, with support $[0, 1]$.
- **Flatness (`flatness`)**: A value near 1 means that the nodule is not flat. Continuous, with support $[0, 1]$ and a point mass at 0.
- **Solidity (`solidity`)**: The number of voxels in the nodule divided by the number in its convex hull. A low value indicates an irregular outline. Continuous, with support $(0, 1]$.
- **Compactness (`compactness`)**: $V^{2/3}/S$, for volume V and surface area S. A high value means little surface area relative to volume. Continuous, with support $(0, \infty)$.

Secondary model: primary features and semantic ratings These are ordinal ratings from 1 to 5, given by the same radiologists who rate malignancy. The secondary model adds them to the five geometric features above. Since each is a median across readers, all five are discrete (ordinal), with support $\{1, 1.5, 2, \ldots, 4.5, 5\}$.

- **Spiculation (`spiculation`)**: Spiky extensions from the edge, from none to marked.
- **Lobulation (`lobulation`)**: Rounded bulges along the edge, from none to marked.
- **Margin (`margin`)**: Definition of the edge, from poorly defined to sharp.
- **Texture (`texture`)**: Internal density, from non-solid to solid.

- **Sphericity (`sphericity`)**: Overall roundness, from linear to round.

The data are split by patient into a training set (609 patients, 1809 nodules) and a test set (261 patients, 828 nodules). Flatness is exactly 0 for 45 training nodules whose outlines lie in a single plane.

2.1 Summary Statistics

All summary statistics are computed on the training set. Table 1 gives the unstandardised values of the predictors, and Table 2 breaks them down by median malignancy grade.

Model	Variable	Mean	Minimum	Maximum	Std. Dev.
Primary	**Log Volume**	5.118	2.442	10.345	1.394
	Elongation	0.753	0.306	0.988	0.121
	Flatness	0.555	0.000	0.921	0.164
	Solidity	0.795	0.228	1.000	0.106
	Compactness	0.177	0.058	0.312	0.044
Secondary	**Spiculation**	1.476	1.000	5.000	0.889
	Lobulation	1.572	1.000	5.000	0.883
	Margin	3.899	1.000	5.000	1.138
	Texture	4.363	1.000	5.000	1.202
	Sphericity	3.789	1.000	5.000	0.826

Table 1: Overall Summary Statistics ($n = 1809$ training nodules)

Model	Variable	Grade 1 (238)	Grade 2 (383)	Grade 3 (804)	Grade 4 (233)	Grade 5 (150)
Primary	**Log Volume**	4.595 (1.003)	4.495 (0.947)	4.761 (0.979)	6.230 (1.264)	7.724 (1.060)
	Elongation	0.749 (0.136)	0.733 (0.124)	0.757 (0.117)	0.769 (0.115)	0.767 (0.108)
	Flatness	0.591 (0.150)	0.514 (0.173)	0.546 (0.167)	0.578 (0.152)	0.606 (0.133)
	Solidity	0.841 (0.092)	0.812 (0.092)	0.808 (0.093)	0.743 (0.105)	0.683 (0.122)
	Compactness	0.183 (0.049)	0.187 (0.043)	0.183 (0.041)	0.156 (0.038)	0.136 (0.031)
Secondary	**Spiculation**	1.174 (0.641)	1.205 (0.651)	1.390 (0.757)	1.803 (0.940)	2.593 (1.256)
	Lobulation	1.273 (0.711)	1.303 (0.733)	1.506 (0.818)	1.899 (0.846)	2.583 (1.004)
	Margin	4.660 (0.794)	3.987 (1.145)	3.744 (1.166)	3.644 (1.068)	3.700 (1.012)
	Texture	4.878 (0.555)	4.355 (1.248)	4.180 (1.334)	4.311 (1.146)	4.620 (0.866)
	Sphericity	4.034 (0.921)	3.819 (0.872)	3.775 (0.785)	3.616 (0.781)	3.670 (0.748)

Table 2: Mean (Std. Dev.) by Median Malignancy Grade (number of training nodules in brackets)

Figure 1 shows that grade 3 is by far the most common response. Figure 2 shows that log volume, solidity and compactness change mainly at grades 4 and 5, whereas elongation

and flatness vary little across grades. Readers often disagree: Fig. 3 shows the difference between the highest and lowest rating on nodules with several readers. Figure 4 gives the correlations between the predictors.

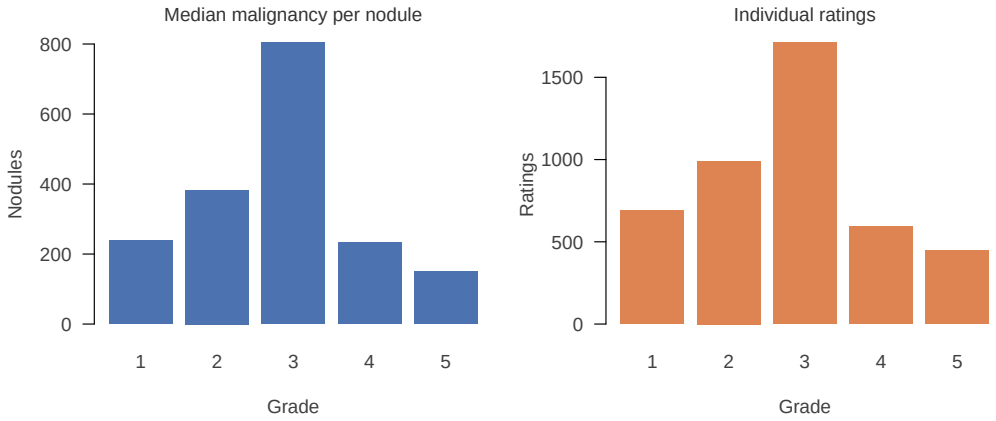


Figure 1: Distribution of the Median Grade and of Individual Ratings (Training Set)

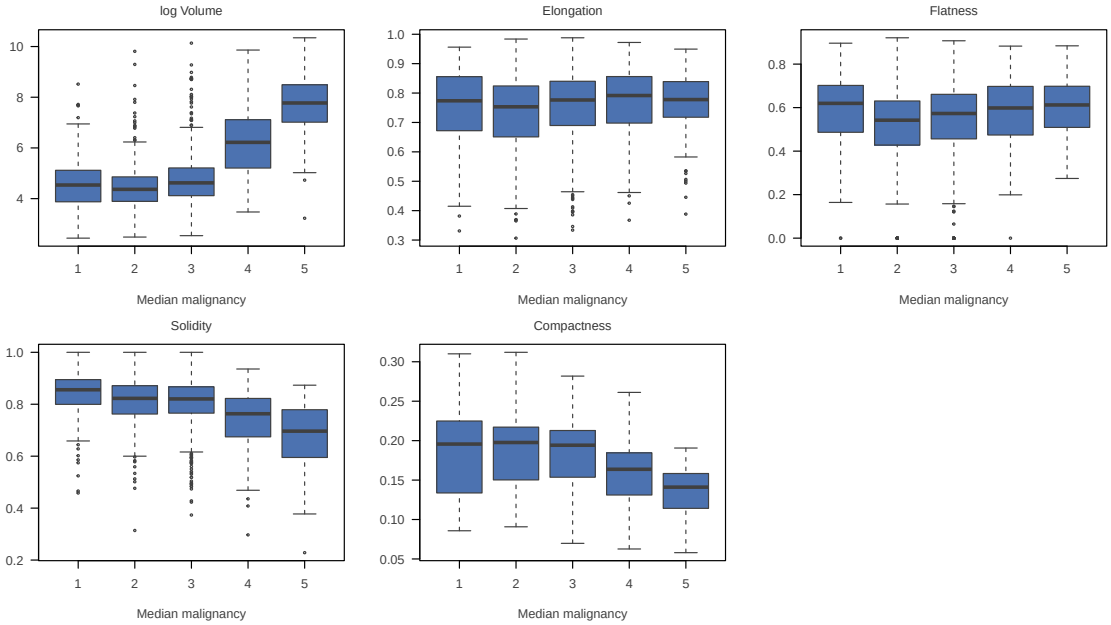


Figure 2: Primary Features by Median Malignancy Grade (Training Set)

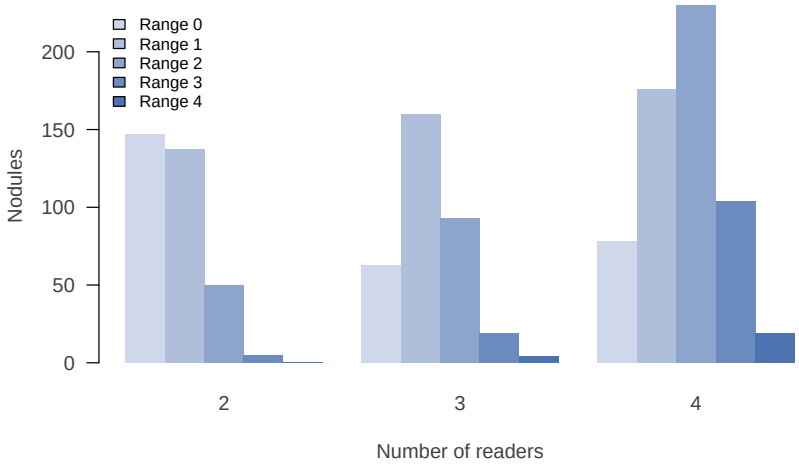


Figure 3: Range of Ratings per Nodule, by Number of Readers (Training Set)

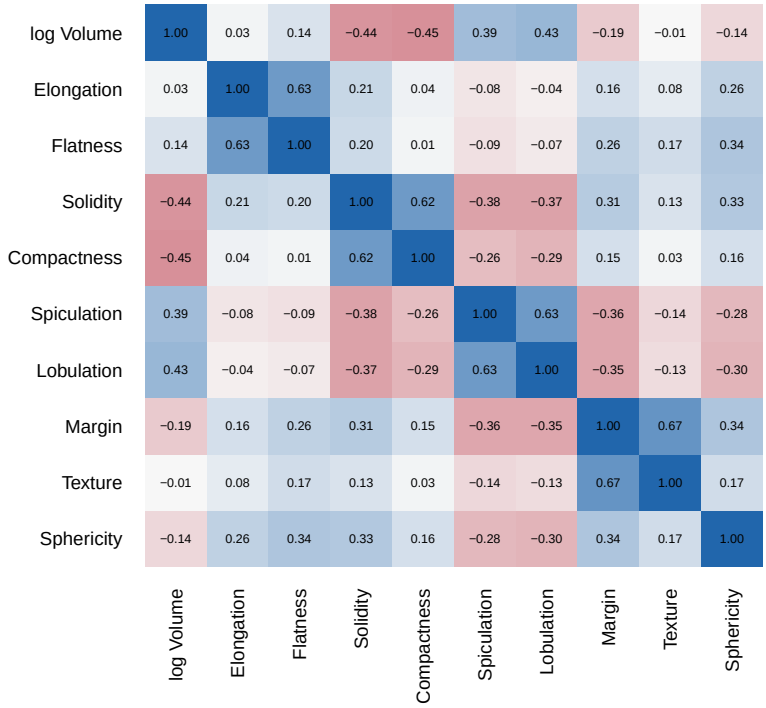


Figure 4: Spearman Correlations Between the Predictors (Training Set)

2.2 Data Source

The dataset used in this study is the LIDC-IDRI database[1], hosted by The Cancer Imaging Archive at LIDC-IDRI (TCIA). The full image collection is about 133 GB, but the radiologists' annotations are bundled, already parsed, with the Python library `pylidc`. Every quantity used in this project is computed from these annotations with `pylidc` version 0.2.3, so no CT images were downloaded.

3 Model Specification

3.1 Logistic Regression Model (Naive Proportional Odds)

Methodology Let $Y_i \in \{1, \dots, 5\}$ be the median malignancy grade of nodule i, and let x_i be its vector of predictors. Since Y_i is ordered, we model its cumulative probabilities with a proportional odds model (an ordinal logistic regression):

$$\operatorname{logit} P(Y_i \le k \mid x_i) = \alpha_k - x_i^{\mathsf{T}}\beta, \qquad k = 1, \dots, 4,$$

where $\alpha_1 < \cdots < \alpha_4$ are thresholds. A positive β_j means that a larger x_{ij} shifts a nodule towards higher grades. The odds of being above any given grade are multiplied by e^{β_j} for each one-unit increase in x_{ij}, and this factor is assumed to be the same at every threshold. The model is *naive* because it takes the median as the true grade and ignores disagreement between readers.

Writing $F(t) = e^t/(1 + e^t)$ for the logistic distribution function, with $\alpha_0 = -\infty$ and $\alpha_5 = \infty$, and treating nodules as independent, the likelihood is

$$L(\alpha, \beta) = \prod_{i=1}^{n} \prod_{k=1}^{5} \left[F(\alpha_k - x_i^{\mathsf{T}}\beta) - F(\alpha_{k-1} - x_i^{\mathsf{T}}\beta)\right]^{\mathbf{1}\{Y_i = k\}}.$$

The models are fitted by maximising L on the training set with `polr`. All predictors are standardised, so one unit is one standard deviation. Confidence intervals for the odds ratios are profile-likelihood intervals, obtained by inverting the likelihood ratio test of $H_0 \colon \beta_j = b$ using the χ^2_1 limit given by Wilks' theorem, and then exponentiating the endpoints.

Wilks' theorem states that, under regularity conditions and if H_0 is true, the likelihood ratio statistic satisfies

$$\Lambda = 2\Big[\sup_{\theta \in \Theta} \ell(\theta) - \sup_{\theta \in \Theta_0} \ell(\theta)\Big] \xrightarrow{d} \chi^2_r$$

as $n \to \infty$, where ℓ is the log-likelihood and r is the number of parameters fixed by H_0. This is a large-sample result. Here $r = 1$, so the approximate 95% interval for β_j is the set of values b with $\Lambda(b) \le \chi^2_{1,0.95} = 3.841$.

Primary Model The fitted model is

$$\begin{aligned}\operatorname{logit} \hat{P}(Y_i \le k \mid x_i) = \hat{\alpha}_k - \big(&1.306 \cdot \texttt{log_volume} + 0.333 \cdot \texttt{elongation} \\ &- 0.178 \cdot \texttt{flatness} - 0.327 \cdot \texttt{solidity} \\ &+ 0.048 \cdot \texttt{compactness}\big),\end{aligned}$$

with $\hat{\alpha} = (-2.456, -1.068, 1.602, 3.291)$ and maximised log-likelihood -2193.908. The estimates are given in Table 3.

Predictor	$\hat{\beta}$	**Std. Error**	**Odds Ratio**	95% **CI**
`log_volume`	1.306	0.064	3.693	(3.259, 4.193)
`elongation`	0.333	0.055	1.396	(1.254, 1.554)
`flatness`	-0.178	0.053	0.837	(0.754, 0.929)
`solidity`	-0.327	0.065	0.721	(0.635, 0.818)
`compactness`	0.048	0.059	1.049	(0.935, 1.177)

Table 3: Primary Proportional Odds Model

Interpretation Each odds ratio is for a one standard deviation increase in that predictor, holding the others constant.

- **Size:** Log volume has by far the largest effect. One standard deviation, roughly a fourfold increase in volume, multiplies the odds of a higher malignancy grade by 3.69.
- **Solidity:** A more solid nodule has lower odds of a higher grade (odds ratio 0.72), so irregular outlines are rated as more suspicious.
- **Elongation and Flatness:** Elongation is close to 1 for a nodule that is not elongated, so an odds ratio of 1.40 means that more elongated nodules have lower odds of a higher grade. Flatness works in the other direction: its odds ratio of 0.84 means that flatter nodules have higher odds.
- **Compactness:** The confidence interval contains 1, so the data are compatible with no effect once the other predictors are included.

Figure 5 shows the fitted probabilities as size increases, with the other predictors at their means.

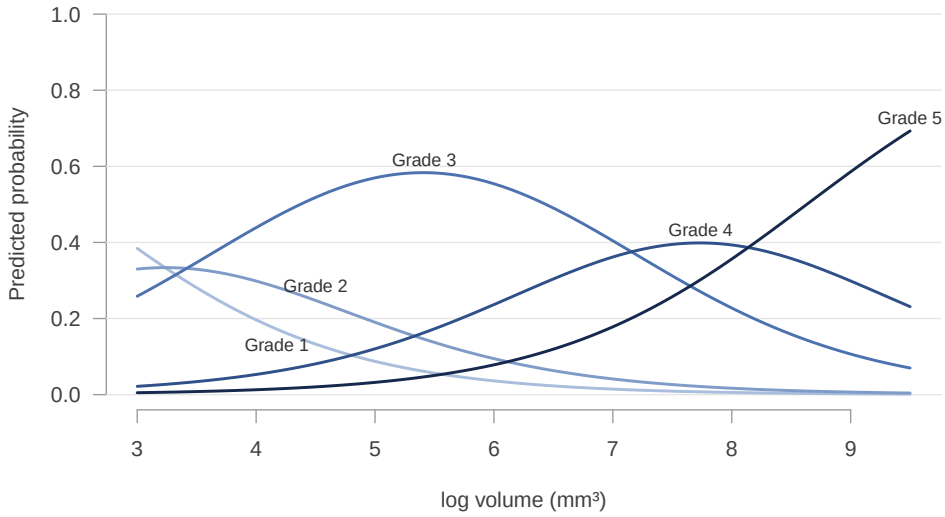


Figure 5: Predicted Probability of Each Grade Against Log Volume, Primary Model

Secondary Model The geometric features only describe the outline of a nodule, whereas radiologists also judge its appearance on the scan. We therefore add the semantic ratings to measure how much this assessment explains beyond geometry: spiculation, lobulation and margin describe the edge in terms that radiologists themselves use, texture records internal density, which the outlines cannot capture, and sphericity gives a reader's view of overall shape. Adding these ratings gives

$$\begin{aligned}\operatorname{logit} \hat{P}(Y_i \le k \mid x_i) = \hat{\alpha}_k - \big(&1.183 \cdot \texttt{log_volume} + 0.321 \cdot \texttt{elongation} \\ &- 0.075 \cdot \texttt{flatness} - 0.158 \cdot \texttt{solidity} \\ &+ 0.008 \cdot \texttt{compactness} + 0.329 \cdot \texttt{spiculation} \\ &+ 0.092 \cdot \texttt{lobulation} - 0.279 \cdot \texttt{margin} \\ &+ 0.067 \cdot \texttt{texture} + 0.014 \cdot \texttt{sphericity}\big),\end{aligned}$$

with $\hat{\alpha} = (-2.518, -1.098, 1.661, 3.404)$ and maximised log-likelihood -2150.257. The estimates are given in Table 4.

Predictor	$\hat{\beta}$	Std. Error	Odds Ratio	95% CI
log_volume	1.183	0.067	3.264	(2.865, 3.724)
elongation	0.321	0.055	1.378	(1.237, 1.537)
flatness	-0.075	0.055	0.928	(0.832, 1.034)
solidity	-0.158	0.069	0.854	(0.746, 0.977)
compactness	0.008	0.059	1.008	(0.898, 1.132)
spiculation	0.329	0.066	1.390	(1.221, 1.584)
lobulation	0.092	0.065	1.097	(0.965, 1.247)
margin	-0.279	0.074	0.757	(0.655, 0.874)
texture	0.067	0.067	1.070	(0.939, 1.219)
sphericity	0.014	0.050	1.014	(0.919, 1.120)

Table 4: Secondary Proportional Odds Model

Interpretation

- **Semantic Ratings:** More spiculated nodules have higher odds of a higher grade (odds ratio 1.39), and nodules with sharper margins have lower odds (0.76). The confidence intervals for lobulation, texture and sphericity contain 1.
- **Geometric Features:** Size remains the largest effect (odds ratio 3.26), and elongation barely changes (1.38). The solidity effect moves towards 1 (0.85), and the interval for flatness now contains 1, which suggests that the ratings capture some of the same information about shape.
- **Circularity:** The semantic ratings come from the same radiologists who rate malignancy, so these associations may partly reflect a reader's overall impression of a nodule.

Figure 6 compares the odds ratios of all the models, including the EM estimates of the next section, which have no confidence intervals.

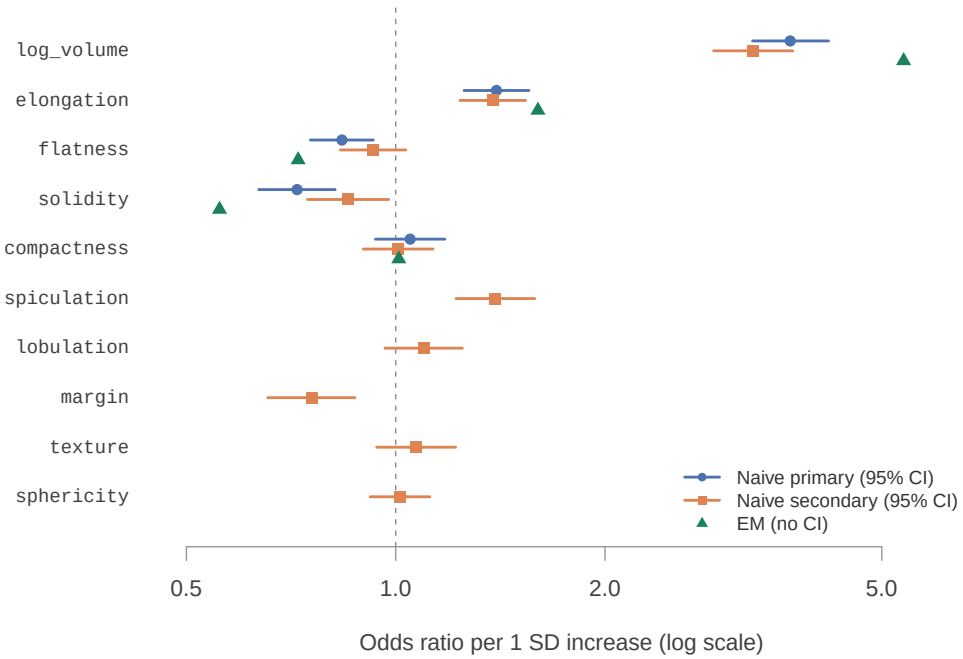


Figure 6: Odds Ratios per Standard Deviation Increase

3.2 Latent True-Grade Model (EM Algorithm)

Methodology The naive model treats the median rating as the true grade. Instead, we now treat the true grade $Z_i \in \{1, \ldots, 5\}$ of nodule i as an unobserved variable, and the readers' ratings R_{ij} as noisy measurements of it. The model has two parts:

- **Structural Model:** The proportional odds model now describes the latent grade, using the primary features x_i:

$$p_{ik}(\alpha, \beta) = P(Z_i = k \mid x_i) = F(\alpha_k - x_i^\mathsf{T}\beta) - F(\alpha_{k-1} - x_i^\mathsf{T}\beta).$$

- **Measurement Model:** A reader rates a nodule of true grade k as r with probability $P(R_{ij} = r \mid Z_i = k) = \pi_{kr}$, where $\sum_r \pi_{kr} = 1$. All readers share this 5×5 matrix, because reader identities are not consistent across scans. Ratings are assumed to be conditionally independent given Z_i.

Let $\Psi = (\alpha, \beta, \pi)$, and let $C_{ir} = \#\{j : R_{ij} = r\}$ be the number of readers who gave nodule i the rating r. The ratings are the observed (*incomplete*) data, and their log-likelihood is

$$\ell(\Psi) = \sum_{i=1}^{n} \log \sum_{k=1}^{5} p_{ik}(\alpha, \beta) \prod_{r=1}^{5} \pi_{kr}^{C_{ir}}.$$

The sum inside the logarithm makes this hard to maximise directly. If Z_i were observed,

the *complete* log-likelihood would separate into simpler terms:

$$\ell_c(\Psi) = \sum_{i=1}^{n} \sum_{k=1}^{5} \mathbf{1}\{Z_i = k\} \Big[\log p_{ik}(\alpha, \beta) + \sum_{r=1}^{5} C_{ir} \log \pi_{kr} \Big].$$

We therefore estimate Ψ with the EM algorithm.

Core Steps Starting from an initial guess $\Psi^{(0)}$, each iteration $t = 0, 1, 2, \ldots$ does the following.

1. **E-step:** Compute the posterior probability of each true grade given the ratings and the current parameters:

$$w_{ik}^{(t)} = P\big(Z_i = k \mid R_i, x_i, \Psi^{(t)}\big) = \frac{p_{ik}\big(\alpha^{(t)}, \beta^{(t)}\big) \prod_r \big(\pi_{kr}^{(t)}\big)^{C_{ir}}}{\sum_{l=1}^{5} p_{il}\big(\alpha^{(t)}, \beta^{(t)}\big) \prod_r \big(\pi_{lr}^{(t)}\big)^{C_{ir}}}.$$

   Taking the expectation of ℓ_c then replaces $\mathbf{1}\{Z_i = k\}$ with $w_{ik}^{(t)}$, which gives the Q-function:

$$Q\big(\Psi \mid \Psi^{(t)}\big) = \sum_{i=1}^{n} \sum_{k=1}^{5} w_{ik}^{(t)} \log p_{ik}(\alpha, \beta) + \sum_{i=1}^{n} \sum_{k=1}^{5} \sum_{r=1}^{5} w_{ik}^{(t)} C_{ir} \log \pi_{kr}.$$

2. **M-step:** Set $\Psi^{(t+1)} = \operatorname{argmax}_{\Psi} Q(\Psi \mid \Psi^{(t)})$. The two terms of Q involve different parameters, so they are maximised separately:

   - The update for π has a closed form, found using a Lagrange multiplier for the constraint $\sum_r \pi_{kr} = 1$:

$$\pi_{kr}^{(t+1)} = \frac{\sum_i w_{ik}^{(t)} C_{ir}}{\sum_i \sum_s w_{ik}^{(t)} C_{is}}.$$

   - The update for (α, β) has no closed form. It is a proportional odds fit in which each nodule appears once for every grade k, with weight $w_{ik}^{(t)}$, and it is computed with `polr`.

3. **Stopping Criterion:** Stop when $\big|\ell(\Psi^{(t+1)}) - \ell(\Psi^{(t)})\big| < \epsilon$ for a small threshold ϵ, and take $\hat{\Psi} = \Psi^{(t+1)}$.

EM never decreases the observed log-likelihood, that is, $\ell(\Psi^{(t+1)}) \geq \ell(\Psi^{(t)})$, but it may converge to a local maximum rather than the global one. The result can therefore depend on $\Psi^{(0)}$.

Estimates The algorithm is run on the training set from six starting values, the weights implied by the median rating and five random perturbations of them. All six reach the

same log-likelihood, -5190.172, so we find no sign of a local maximum. The fitted structural model is

$$\begin{aligned}\operatorname{logit} \hat{P}(Z_i \le k \mid x_i) = \hat{\alpha}_k - \big(&1.682 \cdot \texttt{log_volume} + 0.471 \cdot \texttt{elongation} \\ &- 0.323 \cdot \texttt{flatness} - 0.583 \cdot \texttt{solidity} \\ &+ 0.010 \cdot \texttt{compactness}\big),\end{aligned}$$

with $\hat{\alpha} = (-2.983, -0.494, 1.968, 4.024)$. Table 5 compares the odds ratios with the naive primary model, and Table 6 gives the estimated matrix $\hat{\pi}$.

Predictor	$\hat{\beta}$ **(EM)**	**Odds Ratio (EM)**	**Odds Ratio (Naive)**
`log_volume`	1.682	5.375	3.693
`elongation`	0.471	1.601	1.396
`flatness`	-0.323	0.724	0.837
`solidity`	-0.583	0.558	0.721
`compactness`	0.010	1.010	1.049

Table 5: EM Estimates of the Structural Model

Latent Grade	**Rated 1**	**Rated 2**	**Rated 3**	**Rated 4**	**Rated 5**
1	0.956	0.044	0.000	0.000	0.000
2	0.103	0.395	0.499	0.002	0.000
3	0.035	0.255	0.541	0.169	0.000
4	0.000	0.092	0.313	0.397	0.197
5	0.004	0.000	0.099	0.194	0.703

Table 6: Estimated Rating Probabilities $\hat{\pi}_{kr}$

Figure 7 shows $\hat{\pi}$ as a heatmap, and Fig. 8 shows the observed-data log-likelihood rising at every iteration, as expected for EM.

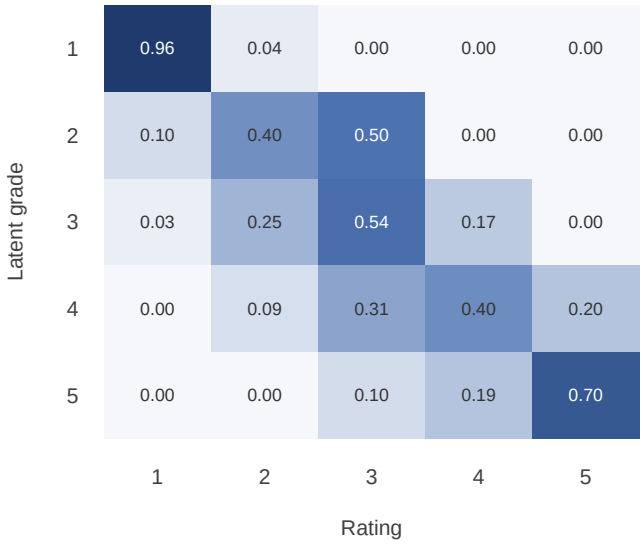


Figure 7: Estimated Rating Probabilities $\hat{\pi}_{kr}$

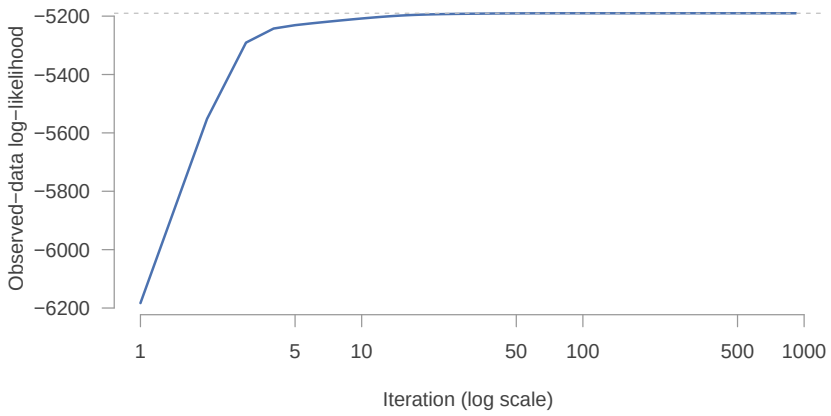


Figure 8: Log-Likelihood Across EM Iterations

Interpretation

- **Structural Model:** The directions of the four main effects match the naive model. Each of these odds ratios is further from 1 under EM, and size remains the largest effect. Compactness is again close to 1. EM does not provide standard errors directly, so these differences are not yet tested.
- **Latent Grades:** Latent grades 1, 3, 4 and 5 are most often rated as themselves, whereas latent grade 2 is rated 3 more often than 2. The latent grades should therefore be read as ordered groups rather than as exact matches to the rating scale.

4 Hypothesis Testing

To formally evaluate our models, we conduct a series of hypothesis tests at the $\alpha = 0.05$ level, all on the training set. All of them are likelihood ratio tests. For a full model M_1 and a restricted model $M_0 \subset M_1$, with maximised log-likelihoods $\hat{\ell}_1$ and $\hat{\ell}_0$, the test statistic is

$$\Lambda = 2\big(\hat{\ell}_1 - \hat{\ell}_0\big).$$

By the Neyman–Pearson lemma, the likelihood ratio test is most powerful when both hypotheses are simple. Our hypotheses are composite, so we rely on Wilks' theorem, under which Λ is approximately χ^2_r under H_0 for large samples, where r is the number of restrictions imposed by M_0. The statistics are computed before rounding, so the last digit may differ from the rounded log-likelihoods shown.

Finding $\hat{\ell}_1$ and $\hat{\ell}_0$ Each is the value of the log-likelihood at its maximum, so every test needs two fits, one over the full parameter space Θ and one over the subset Θ_0 allowed by H_0:

$$\hat{\ell}_1 = \max_{\theta \in \Theta} \ell(\theta), \qquad \hat{\ell}_0 = \max_{\theta \in \Theta_0} \ell(\theta).$$

Setting a coefficient to zero is the same as removing that predictor, and all the remaining parameters are re-estimated in the restricted fit.

For the naive models, the equations $\nabla \ell(\alpha, \beta) = 0$ have no closed-form solution, so ℓ is maximised iteratively with BFGS, a quasi-Newton method. Writing $\theta = (\alpha, \beta)$, each step is

$$\theta^{(t+1)} = \theta^{(t)} + s_t B_t \nabla \ell\big(\theta^{(t)}\big),$$

where s_t is a step size and B_t approximates the inverse of the negative Hessian, updated from the change in the gradient between steps. Since ℓ is concave in (α, β), this reaches the global maximum.

For the EM model, $\ell(\Psi)$ is maximised with the algorithm of the previous section, in which the update for (α, β) in each M-step also uses BFGS. We then evaluate ℓ at the converged parameters. This likelihood need not be concave, so a fit may stop at a local maximum. We therefore fit the full model from six starting values, and each drop-one model from two, keeping the largest log-likelihood. The naive and EM log-likelihoods are based on different data, the median grade and the individual ratings, so they are only compared within each family.

4.1 Overall Model Significance

This test evaluates whether a naive model as a whole explains the malignancy grade better than a model with only the thresholds.

- **Null Hypothesis (H_0):** All coefficients are zero, so the predictors have no association with the malignancy grade.
- **Alternative Hypothesis (H_1):** At least one $\beta_j \neq 0$.

The primary model has 5 coefficients and the secondary model has 10, so Λ is compared with the χ^2_5 and χ^2_{10} distributions respectively.

Calculation and Inference The thresholds-only model has $\hat{\ell}_0 = -2578.524$. For the primary model,

$$\Lambda = 2\big(-2193.908 - (-2578.524)\big) = 769.231,$$

and for the secondary model,

$$\Lambda = 2\big(-2150.257 - (-2578.524)\big) = 856.534.$$

Both p-values are strictly less than 0.001, so we reject the null hypothesis for both models. The predictors are jointly associated with the malignancy grade.

4.2 Individual Parameter Significance

While the overall test evaluates a model as a whole, a drop-one test evaluates the contribution of each predictor given the others.

- **Null Hypothesis (H_0):** $\beta_j = 0$. The j-th predictor has no association with the malignancy grade once the other predictors are included.
- **Alternative Hypothesis (H_1):** $\beta_j \neq 0$.

The restricted model is the full model refitted without the j-th predictor, so $r = 1$ and Λ is compared with the χ^2_1 distribution. The full models have $\hat{\ell}_1 = -2193.908$ (primary) and $\hat{\ell}_1 = -2150.257$ (secondary).

Predictor	Primary Model			Secondary Model		
	$\hat{\ell}_0$	Λ	p-value	$\hat{\ell}_0$	Λ	p-value
log_volume	-2429.043	470.270	< 0.001	-2320.867	341.220	< 0.001
elongation	-2212.687	37.556	< 0.001	-2167.237	33.961	< 0.001
flatness	-2199.565	11.314	< 0.001	-2151.176	1.838	0.175
solidity	-2206.846	25.874	< 0.001	-2152.895	5.276	0.022
compactness	-2194.241	0.665	0.415	-2150.266	0.018	0.893
spiculation	–	–	–	-2162.799	25.084	< 0.001
lobulation	–	–	–	-2151.258	2.002	0.157
margin	–	–	–	-2157.508	14.501	< 0.001
texture	–	–	–	-2150.770	1.026	0.311
sphericity	–	–	–	-2150.297	0.080	0.777

Table 7: Drop-One Likelihood Ratio Tests for the Naive Models

Inference As shown in Table 7, we reject the null hypothesis for log volume, elongation, flatness and solidity in the primary model, and log volume has by far the largest statistic. We fail to reject it for compactness. In the secondary model, we reject it for log volume, elongation, solidity, spiculation and margin. Flatness is no longer significant once the semantic ratings are included. These tests are not adjusted for multiple comparisons.

4.3 Joint Significance of the Semantic Ratings

To justify including the semantic ratings, we test their joint significance against the primary model, which is nested in the secondary model.

- **Null Hypothesis (H_0):** The coefficients of spiculation, lobulation, margin, texture and sphericity are all zero. The ratings provide no additional explanatory power beyond the geometric features.
- **Alternative Hypothesis (H_1):** At least one of these coefficients is non-zero.

The secondary model adds 5 coefficients, so $r = 5$.

Calculation and Inference Evaluating the statistic gives

$$\Lambda = 2\big(-2150.257 - (-2193.908)\big) = 87.303,$$

with a p-value strictly less than 0.001 under the χ^2_5 distribution. We therefore reject the null hypothesis. The semantic ratings improve the model, although this is partly circular

because they come from the same radiologists who rate malignancy.

4.4 Significance of the Features in the EM Model

The same tests apply to the EM model, using the observed-data log-likelihood $\ell(\Psi)$ rather than the complete-data one. Each restricted model is refitted by EM. The full model has $\hat{\ell}_1 = -5190.172$.

Overall Significance

- **Null Hypothesis (H_0):** $\beta = 0$. The features are not informative about the latent grade beyond the readers' ratings, so $p_{ik} = p_k$ for every nodule.
- **Alternative Hypothesis (H_1):** At least one $\beta_j \neq 0$.

The restricted model has $\hat{\ell}_0 = -5606.396$, which gives

$$\Lambda = 2\big(-5190.172 - (-5606.396)\big) = 832.449.$$

With $r = 5$, the p-value is strictly less than 0.001, so we reject the null hypothesis. The features are informative about the latent grade.

Individual Parameter Significance We then test $H_0 \colon \beta_j = 0$ against $H_1 \colon \beta_j \neq 0$ for each feature, with $r = 1$.

Predictor	$\hat{\ell}_0$	Λ	**p-value**
`log_volume`	-5424.701	469.058	< 0.001
`elongation`	-5210.130	39.917	< 0.001
`flatness`	-5200.096	19.848	< 0.001
`solidity`	-5212.612	44.880	< 0.001
`compactness`	-5190.181	0.018	0.894

Table 8: Drop-One Likelihood Ratio Tests for the EM Model

Inference As shown in Table 8, we reject the null hypothesis for every feature except compactness. This agrees with the primary naive model: the same four features are significant, and log volume again has the largest statistic.

4.5 Proportional Odds Assumption

The models assume that each coefficient is the same at every threshold. We test this on the naive models, where each threshold has its own coefficient $\beta_j^{(k)}$ under the alternative.

- **Null Hypothesis (H_0):** $\beta_j^{(1)} = \beta_j^{(2)} = \beta_j^{(3)} = \beta_j^{(4)}$. The effect of the j-th predictor is the same at every threshold.
- **Alternative Hypothesis (H_1):** The effect differs for at least one threshold.

We use likelihood ratio tests. The restricted model is the proportional odds model, and the full model lets the coefficients differ by threshold. Freeing one predictor adds 3 parameters, so $r = 3$. Freeing all predictors gives $r = 15$ for the primary model and $r = 30$ for the secondary model.

Calculation and Inference Freeing all predictors in the primary model gives

$$\Lambda = 2\big(-2090.864 - (-2193.908)\big) = 206.089,$$

and in the secondary model it gives

$$\Lambda = 2\big(-2005.144 - (-2150.257)\big) = 290.227.$$

Both p-values are strictly less than 0.001, so we reject the proportional odds assumption for both models. Table 9 gives the test for each predictor.

	Primary Model		Secondary Model	
Predictor	Λ	**p-value**	Λ	**p-value**
`log_volume`	136.258	< 0.001	137.293	< 0.001
`elongation`	7.979	0.046	9.285	0.026
`flatness`	48.578	< 0.001	53.972	< 0.001
`solidity`	26.207	< 0.001	17.305	< 0.001
`compactness`	46.604	< 0.001	40.818	< 0.001
`spiculation`	–	–	4.703	0.195
`lobulation`	–	–	12.084	0.007
`margin`	–	–	76.420	< 0.001
`texture`	–	–	61.760	< 0.001
`sphericity`	–	–	6.303	0.098

Table 9: Likelihood Ratio Tests of the Proportional Odds Assumption ($r = 3$)

Every primary feature violates the assumption at the $\alpha = 0.05$ level. In the secondary

model, only spiculation and sphericity do not. Table 10 shows how the odds ratios of the primary model change across thresholds when all predictors are free.

Predictor	$Y > 1$	$Y > 2$	$Y > 3$	$Y > 4$
`log_volume`	1.857	2.347	4.370	6.266
`elongation`	1.539	1.383	1.372	1.099
`flatness`	0.611	0.866	0.988	1.478
`solidity`	0.548	0.778	0.770	0.701
`compactness`	1.285	1.062	0.774	0.720

Table 10: Odds Ratios at Each Threshold, Primary Model

Figure 9 shows the same odds ratios, with a dotted line at the single proportional odds estimate and a dashed line at 1.

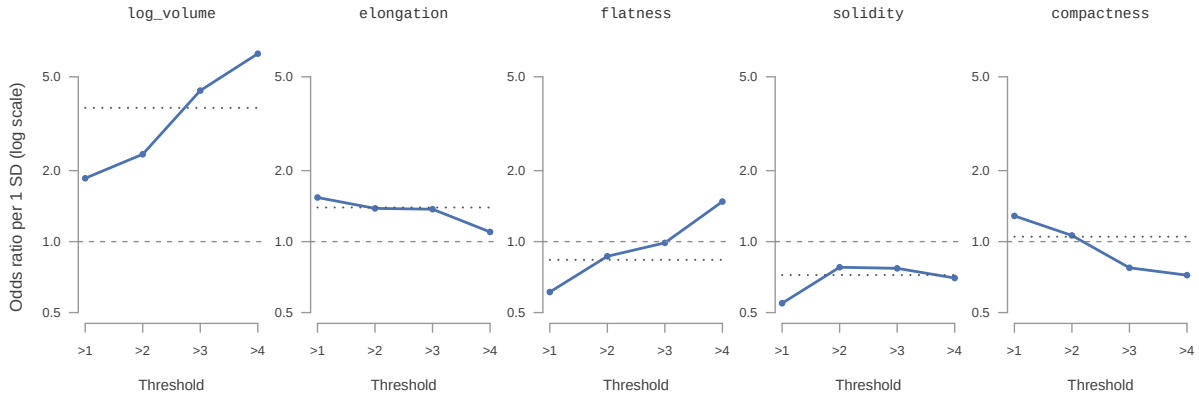


Figure 9: Odds Ratios at Each Threshold, Primary Model

Size matters much more for separating the high grades than the low ones, so its single odds ratio is a summary rather than a constant multiplier. The effects of flatness and compactness change direction across thresholds, so their single odds ratios should not be interpreted on their own. The EM model assumes proportional odds for the latent grade, and we do not test this directly.

5 Model Evaluation

The models are evaluated once on the test set, which contains patients not used for fitting. The coefficients are frozen and the test features are standardised with the training means and standard deviations. One test nodule with missing solidity is excluded, leaving 827 nodules from 261 patients.

5.1 Prediction Errors

Each model gives a probability for each grade, and its prediction is the most likely grade. The EM model's latent grades do not match the rating scale exactly, so we first convert them to rating probabilities:

$$P(R = r \mid x_i) = \sum_{k=1}^{5} p_{ik}(\hat{\alpha}, \hat{\beta})\, \hat{\pi}_{kr}.$$

We compare the prediction $\hat{Y}_i$ with the median grade Y_i using the misclassification rate, the mean absolute error $\text{MAE} = \frac{1}{n}\sum_i |\hat{Y}_i - Y_i|$, and the root mean squared error $\text{RMSE} = \left(\frac{1}{n}\sum_i (\hat{Y}_i - Y_i)^2\right)^{1/2}$. The baseline always predicts grade 3, the most common training grade.

	Error			**Reduction vs. Baseline (%)**		
Model	**Misclass.**	**MAE**	**RMSE**	**Misclass.**	**MAE**	**RMSE**
Baseline	0.512	0.676	1.002	–	–	–
Naive Primary	0.507	0.652	0.974	0.95	3.58	2.81
Naive Secondary	0.486	0.616	0.940	4.96	8.94	6.21
EM	0.510	0.688	1.031	0.24	-1.79	-2.85

Table 11: Test-Set Errors

Model	**Grade 1**	**Grade 2**	**Grade 3**	**Grade 4**	**Grade 5**
Actual Median	89	173	404	114	47
Naive Primary	44	20	676	46	41
Naive Secondary	44	29	659	57	38
EM	65	0	702	0	60

Table 12: Number of Test Nodules Predicted at Each Grade

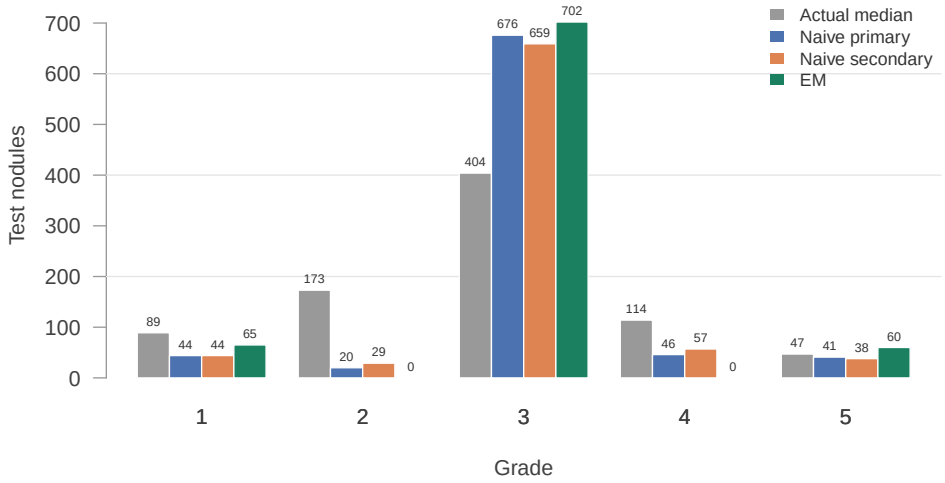


Figure 10: Number of Test Nodules at Each Grade: Actual and Predicted

The naive secondary model has the lowest errors, and the naive primary model improves only slightly on the baseline. The EM model does not improve on the baseline: its misclassification rate is about the same, and its MAE and RMSE are worse. These are point estimates without confidence intervals, so small differences should not be treated as real. For example, the naive primary and EM models classify only 4 and 1 more nodules correctly than the baseline.

5.2 Reader Benchmark

The grades are themselves radiologists' opinions, so no model can be expected to have zero error. To put the errors in context, we compare each model with a single reader. For each reader on a nodule with at least two readers, the target is the median of the *other* readers, with half-grades rounded up. We then measure how far that reader's rating and each model's prediction are from this target. This uses the 581 multi-reader test nodules, on 79.9% of which the readers do not all agree.

Predictor	**MAE Against Other Readers**
Single Reader	0.743
Baseline	0.709
Naive Primary	0.614
Naive Secondary	0.579
EM	0.641

Table 13: Reader Benchmark on Multi-Reader Test Nodules

All the models are closer to the other readers' median than a single reader is. This should

not be overstated, because the constant baseline is too: MAE rewards predictions near the centre of the scale, whereas an individual reader's rating varies. The benchmark therefore shows that much of the error reflects disagreement between readers, not that the models outperform radiologists.

5.3 Shortcomings

- **Small gains over the baseline:** Grade 3 makes up 48.9% of the test set, so always predicting it is already hard to beat. The most likely grade also rarely moves away from it: the naive primary model predicts grade 3 for 81.7% of nodules. Grade 2 is especially rare: it is the most likely grade only when $x_i^\mathsf{T}\hat{\beta}$ lies between -1.765 and -1.596, against -1.595 to 1.909 for grade 3, and grades 1 to 3 have similar mean log volumes in training (4.60, 4.50 and 4.76), so the features do little to separate them.
- **No improvement from EM:** Using the most likely grade, the EM model never predicts grades 2 or 4. In Table 6, latent grades 2 and 3 are both rated 3 most often, so a rating of 3 dominates the middle of the scale. When its predictions leave grade 3, they go straight to 1 or 5, which raises the MAE and RMSE. In addition, the EM model is fitted to individual ratings, whereas these errors are measured against the median, which favours the naive models.
- **Missing calcification:** Calcification is excluded from our models, although calcified nodules are rated much lower (mean grade 1.22 against 3.02). These 87 nodules are 10.5% of the test set but account for about a quarter of the naive primary model's absolute error, with an MAE of 1.64 against 0.54 for the others. This comparison was made after seeing the results.
- **Limited features:** Without the CT images, we have no intensity-based features, and the geometric features only describe an outline drawn by the radiologists.
- **Violated assumptions:** The proportional odds assumption is rejected, so one coefficient per predictor cannot capture effects that differ across thresholds. Nodules from one patient are also treated as independent.
- **Circularity of the best model:** The naive secondary model performs best, but its semantic ratings come from the same radiologists who rate malignancy, and they would not be available without a reader. Its advantage should not be read as evidence that it would help automated prediction.
- **Noisy target:** The labels are radiologists' suspicion rather than confirmed diagnoses, and readers frequently disagree. Good performance here means agreement with expert opinion, not detection of cancer.

- **No measures of uncertainty:** We did not compute confidence intervals for these errors or for the EM coefficients, and we did not run a simulation study. We therefore cannot say whether the differences between models are real, or confirm that the EM estimates recover known parameters.

6 Policy Interpretation

Most lung nodules are benign, and radiologist time is limited in public health systems. We therefore ask whether a model could help prioritise nodules for review. This is a second-reader setting: the model would decide which nodules are reviewed first, not which are reviewed at all, and it would not replace a radiologist.

6.1 A Triage Rule

We flag nodule i for priority review if its predicted probability of a high grade exceeds a threshold:

$$P(Y_i \geq 4 \mid x_i) > \tau.$$

The threshold is chosen on the training set as the largest value reaching 90% sensitivity, and is then applied unchanged to the test set. Sensitivity is the share of nodules with grade at least 4 that are flagged, workload is the share of all nodules flagged, and precision is the share of flagged nodules with grade at least 4. In the test set, 161 of the 827 nodules (19.5%) have grade at least 4. For the EM model, the probability is taken on the rating scale.

Model	τ	Sensitivity	Workload	Precision	Missed
Naive Primary	0.139	0.913	0.470	0.378	14
Naive Secondary	0.146	0.894	0.426	0.409	17
EM	0.147	0.907	0.441	0.400	15

Table 14: Triage Rule on the Test Set

Figure 11 shows the trade-off between sensitivity and workload as τ varies. The three curves almost coincide, so the models order nodules very similarly, and all are well above random flagging.

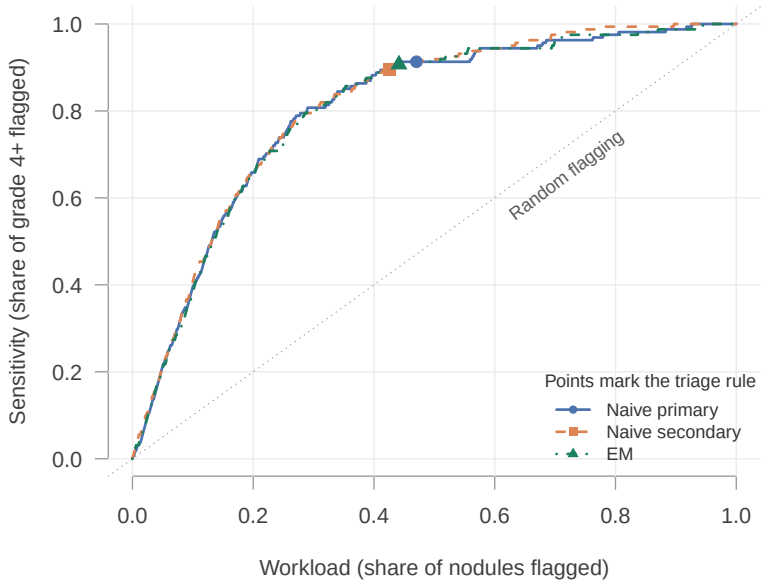


Figure 11: Sensitivity Against Workload on the Test Set

6.2 Recommendation

Under the naive primary rule, a radiologist reviewing the flagged nodules first would see 91.3% of the high-grade nodules within the first 47.0% of cases. About 38% of flagged nodules are high grade, roughly twice the rate in the test set. So although the models predict exact grades poorly, their probabilities are still useful for ordering nodules.

- We recommend the naive primary model as the starting point for a triage tool. It is the simplest model and needs no semantic ratings, and the small differences in Table 14 are not enough to prefer a more complex model.
- The naive secondary model should not be used for triage, as its semantic ratings require the radiologist review that triage is meant to prioritise.
- The rule should never be used to dismiss a nodule. It misses 14 of the 161 high-grade test nodules, and each of these should still receive a routine review.

6.3 Limitations

- **Suspicion, not diagnosis:** LIDC-IDRI malignancy is radiologists' suspicion rather than a pathology-confirmed diagnosis, so the rule predicts expert opinion, not cancer. The Cancer Imaging Archive holds diagnosis data for a small subset of patients, which could be used for external validation.
- **Dependence on drawn outlines:** Our geometric features come from outlines

drawn by radiologists. A working tool would need automatic segmentation, whose performance we have not assessed.

- **Missing features:** We use no intensity-based features, and calcification, a strong sign of a benign nodule, is not in the model.
- **Limited certainty:** The results have no confidence intervals, and the 90% sensitivity target is an illustrative choice. A real deployment would need to set it based on the clinical cost of a missed nodule.
- **Generalisation:** LIDC-IDRI is a research dataset from a small number of centres in the United States, so its use in a different setting, such as Indian public hospitals, is untested.

7 Conclusion

This study modelled radiologists' malignancy ratings of lung nodules in the LIDC-IDRI database. We fitted a proportional odds model to the median rating, first with geometric features and then with semantic ratings added, and a latent-grade model, fitted by the EM algorithm, that treats each nodule's true grade as unobserved.

Size is the dominant predictor. In the primary model, one standard deviation of log volume multiplies the odds of a higher grade by 3.69, and log volume, elongation, flatness and solidity are all significant. Compactness is not significant in any model. The semantic ratings improve the fit, particularly spiculation and margin, although this is partly circular because they come from the same radiologists. The EM model gives the same directions for these effects, with odds ratios further from 1, but we did not assess the uncertainty of those differences.

The proportional odds assumption is rejected for both naive models. The effect of size grows across the thresholds, and the effects of flatness and compactness change direction, so the single odds ratios are only summaries.

On the test set, the models predict exact grades poorly. The naive secondary model does best, with a misclassification rate of 48.6% against 51.2% for always predicting grade 3, and the naive primary model barely improves on this baseline. The EM model does not improve on it, and has a higher MAE and RMSE. Modelling the latent grade therefore gave no predictive benefit here, which suggests that the median rating is an adequate summary for prediction. Much of the remaining error reflects disagreement between readers, as a single reader differs from the other readers by 0.74 grades on average.

The models are more useful for ordering nodules than for assigning grades. A triage rule based on the naive primary model flags 91.3% of high-grade test nodules while flagging

47.0% of all nodules. This supports further study of such a model as an aid to prioritising review, but not as a replacement for a radiologist.

These conclusions are limited by the use of radiologists' suspicion rather than confirmed diagnoses, the exclusion of calcification and of intensity-based features, the violated assumptions, and the absence of confidence intervals for the test results. Future work should add calcification to the models, relax the proportional odds assumption and validate the triage rule against confirmed diagnoses.

A Appendix

A.1 Dataset

The dataset used in this study is the LIDC-IDRI database[1]. It can be accessed through The Cancer Imaging Archive: cancerimagingarchive.net/collection/lidc-idri. The annotations were read with the Python library `pylidc`, version 0.2.3.

A.2 Code

The code used in this project is divided based on functionality:

- Python Code (Data Extraction): Python Code Link
- R Code (Computation and Model Fitting): R Code Link
- R Code (Visualization and Plotting): Plotting Code Link

References

[1] Samuel G. Armato, Geoffrey McLennan, Luc Bidaut, Michael F. McNitt-Gray, Charles R. Meyer, Anthony P. Reeves, et al. The Lung Image Database Consortium (LIDC) and Image Database Resource Initiative (IDRI): A completed reference database of lung nodules on CT scans. *Medical Physics*, 38(2):915–931, 2011.